

3.3 Spring Data Jpa Interfaces and Dynamic Query Methods

• Spring Data JPA:

- Spring Data JPA is a part of the larger **Spring Data family**.
- It builds on top of **JPA**, providing a higher-level and more convenient abstraction for data access.
- **Spring Data JPA** makes it easier to implement **JPA-based repositories** by providing boilerplate code, custom query methods, and various utilities to reduce the amount of code you need to write.

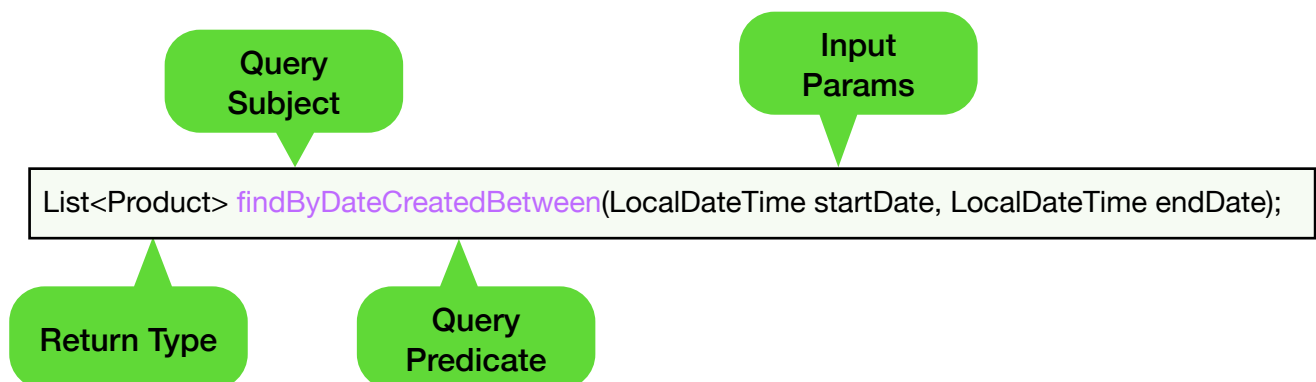


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- **Key features of Spring Data JPA:**

1. **Repository Abstraction:** Provides a **Repository** interface with **methods** for common data access operations.
2. **Custom Query Methods:** Allows defining **custom query methods** by simply declaring **method names**.
3. **Pagination and Sorting:** Offers built-in support for **pagination** and **sorting**.
4. **Query Derivation:** Automatically **generates queries** from **method names**.

- **Rules for Creating Query Methods:**



1. The name of our query method must start with one of the following prefixes: **find...By**, **read...By**, **query...By**, and **get...By**.
 - Examples: **findByName**, **readByName**, **queryByName**, **getByName**
2. If we want to **limit** the **number** of returned **query results**, we can add the **First** or the **Top** keyword before the first "By" word.
 - Examples: **findFirstByName**, **readFirst2ByName**, **findTop10ByName**
3. If we want to select **unique** results, we have to add the "**Distinct**" keyword before the first "By" word.
 - Examples: **findDistinctByName** or **findNameDistinctBy**
4. **Combine property** expression with **AND** and **OR**.
 - Examples: **findByNameOrDescription**, **findByNameAndDescription**
5. Checkout the Official Doc for more info: <https://docs.spring.io/spring-data/jpa/reference/repositories/querykeywords-reference.html>